

PRACHI SINGH

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in Prachi Singh

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SUMMARY

Passionate and skilled Computer Vision & Machine Learning enthusiast with hands-on experience in AI-driven sports analytics and gesture recognition systems. Proficient in Python, C, and C++, with a strong foundation in Data Structures & Algorithms.

WORK EXPERIENCE

Software Intern, Aurionpro Solutions Limited

Apr 2025 - June 2025

- Built an Invoice Reader system to extract key-value pairs from diverse PDF invoices using a hybrid machine learning and rule-based approach.
- Utilized LayoutLMv2 for layout-aware extraction, bypassing OCR with PDF text extraction, and tested spaCy's NER pipeline despite setup challenges.
- Resolved technical issues like model loading and dependency errors, delivering a pipeline with JSON, CSV, and tabular outputs.

Executive Grade Consultant (Intern), Datamatics

May 2026 - July 2026

- Performed functional and API testing of enterprise web applications to ensure reliability and performance.
- Developed automation test scripts using TypeScript and Playwright, reducing repetitive manual testing efforts.
- Analyzed application workflows, identified defects, and collaborated with developers to resolve issues efficiently.
- Validated REST API endpoints and verified application behavior across different test scenarios.

EDUCATION

VIT Bhopal University, Bhopal, Madhya Pradesh

August 2023 - June 2027

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.32/10

Delhi Public School (DPS), Vadodara, Gujarat

April 2022 - May 2023

Class XII-PCM

Percentage: 73%

Delhi Public School (DPS), Bhopal, Madhya Pradesh

April 2021

Class X

Percentage: 90%

PROJECTS

PlayVision: AI Powered Player Performance Tracking

[Link ↗](#)

- Developed PlayVision, an AI-powered football analytics system using YOLOv8, ByteTrack, and SigLIP for real-time player and ball tracking, team identification, and spatial analysis.
- Implemented multi-object tracking with ByteTrack and UMAP for positional clustering, enabling computation of metrics like ball possession, player speed, and territory control.
- Overcame challenges like occlusion and latency, delivering a tactical radar view to enhance coaching, scouting, and fan engagement, with potential for broader league applications.

Aero Gest

[Link ↗](#)

- Developed a virtual mouse system using hand gesture recognition to control cursor movement, clicks, and system settings like volume and brightness, leveraging OpenCV for video capture and MediaPipe for real-time hand tracking.
- Utilized Python libraries (PyAutoGUI, Pycaw, Screen Brightness Control) to map mathematically modeled hand gestures to specific computer commands, ensuring real-time responsiveness and smooth operation.
- Achieved high accuracy and usability across various lighting conditions, offering a touchless, intuitive alternative to touch-based and voice-controlled interfaces for assistive technology and contactless environments.

ADDITIONAL INFORMATION

- Technical Skills:** C, C++, Python, Java, CSV, Data Structures, JSON, OpenCV, SpaCy
- Languages:** English, Hindi
- Volunteering:** VIT Bhopal Annual Convocation Volunteer, Member of the Event Management Committee